CHEMICAL ENGINEERING
COLLEGE OF ENGINEERING
UIC

Department of Chemical Engineering
810 South Clinton Street, 217 CEB
Chicago, IL 60607-4408

AGENDA:
Department of Chemical Engineering
Curriculum Day – 10/16/2018 (CEB 230 C)

9:30 AM	Welcome: Berry and Zdunek	
9:35 AM	Flow Chart of Curriculum: Zdunek	
9:40 AM	Current Curriculum of ChE. Each faculty presents the curriculum for their course:	
	CHE 201	
	CHE 205	
	CHE 210	
	CHE 301	
	CHE 311	
	CHE 312	
10:15	Break	
10:25	Nicole Gesualdo (Communications)	
10:35	CHE 313	
	CHE 321	
	CHE 341	
	CHE 381 / 382	
	CHE 396 / 397	
	CHE 422	
11:30	Curriculum Discussion	All
11:45	Best Practices in Teaching	Wedgewood
11:55	Best Practices in Mentoring UGs for research	Sharma
12:05	Teaching Methodologies	Bilgin
12:15	Chem-E-Car and Student Community	Singh
12:30	Lunch + Discussion on Improving Student Experience	Berry/Nitsche.

Handouts provided for course syllabi, PEO's and student outcomes (2014 and new SO's)

PEO's and student Outcomes discussed here
AZ

Phone (312) 996-2342 • Fax (312) 996-0808 • vikasb@uic.edu

Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

Chemical Engineering Department
College of Engineering

Topic: Mapping of ABET Student Learning Outcomes to ChE Courses
Date: September 7, 2018

Course Number	
Faculty Name	

Please identify your course contributions to the following ABET student outcomes using the following: I: Introduce, R: Reinforce, E: Emphasize

		I	R	E
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			
(3)	an ability to communicate effectively with a range of audiences.			
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			

Introduced (I): Students are not expected to be familiar with the content or skill at this level. Instructions and learning activities focus on basic level of knowledge.
Reinforced (R): students have been introduced before to the content or skill. Instructions and learning activities continue to build upon previous competencies with increased complexity.
Emphasized (E): Students are expected to possess an advanced level of knowledge, skill, or competency at this level. Students are asked to demonstrate their learning on the outcome through homework, projects, tests, etc and are provided a feedback for continuous improvement.

ABET 2014 Criterion and Student Outcomes

SO	Description	ABET	PEO
1	Ability to apply knowledge of mathematics, science and engineering	a	1
2	Ability to identify, formulate and solve engineering problems	e	1
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	k	1
4	Ability to design experiments	b	1
5	Ability to conduct experiments	b	1
6	Ability to analyze and interpret data	b	1
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	c	1
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	h	2
9	Knowledge of contemporary issues	j	2
10	Recognition of the need for and ability to engage in life-long learning	i	4
11	Ability to function on multidisciplinary teams	d	3
12	Ability to communicate effectively	g	3
13	Understanding of professional and ethical responsibilities	f	3

SHORTENED SYLLABUS: CHE 201 THERMODYNAMICS

Course Description: 1st Law of thermodynamics; work, heat and conversion of energy in closed and open systems, including engineering components, power and refrigeration cycles. Engineering problem-solving methodology, pure substance properties, property tables, equations of state, theory of gases and other states of matter. 2nd Law of thermodynamics; entropy, open and closed system entropy balances and equilibrium, with applications. Fundamental thermodynamic functions, Maxwell relations, Clapeyron equation.

Text: *Fundamentals of Engineering Thermodynamics*, 8th ed., Moran, Shapiro, Boettner and Bailey 2014

Aims, Scope, Level

This course is the first in the thermodynamics sequence and is designed to give the student a firm grounding in energy and entropy concepts. Later courses will deal in more detail with phase and reaction equilibrium. Knowledge of thermodynamics is needed in subsequent undergraduate chemical engineering courses.

It is assumed that the students in the course have the necessary background in elementary mathematics, and this mathematical background will be used. Also, students should understand mechanics which is usually taught in the first college physics course.
By the end of the term, the students will understand energy and entropy concepts for closed systems and open systems (control volumes). Student will be able to apply engineering methodology to a variety of situations in order to solve thermodynamic problems.

Exams, Homework and Grading

Two exams (20% each)	40%
Final exam	30%
Quizzes (5-6)	15%
Homework problems (13 sets, 5 problems each)	15%

PollEverywhere quizzes are given periodically during semester (not graded)

Detailed Course Outline

Introduction	Energy Usage in the 21st Century Presentation History of Thermodynamics Engineering Ethics (AICHE Code and videos)
Fundamentals	Definitions (System, State, Property), Units, Unit Conversion, Pressure, Temperature, Specific Volume Dimensional Homogeneity Engineering Methodology
1st Law: Closed Systems	Kinetic, Potential, Internal Energy Energy Transfer (Heat & Work) Closed System Energy & Energy Rate Balance Quasi-equilibrium, Polytropic processes Thermodynamic cycles (Power, Refrigeration, Heat Pump)
Pure Substance Properties	Definitions, Property Tables and Calculations Dependent Properties, Reference State Phase Diagrams (T-v, P-v), Quality Using Property Tables in Closed System Energy Balances Linear and Double Interpolation Steam Table Process Overview Property Approximations Equations of State, Compressibility Charts Specific Heats and Heat Capacity Ideal Gas Model
1st Law: Open Systems	Control Volumes, Conservation of Mass, Mass Flow Rate, Volumetric Flowrate Mass Balance and Mass Rate Balance Control Volume Energy & Energy Rate Balance Nozzles/Diffuses, Turbines, Pumps/Compressors, Heat Exchangers, Throttle Valves Property Variability System Integration of Components Transient Analysis
2nd Law	Entropy Statement, Process Spontaneity, Dissymmetry of Nature, Thermal Reservoirs, Clausius Statement, Kelvin-Planck Statement, 2nd Law Cycle Analysis (Power, Refrigeration/Heat Pump), Carnot Corollaries, Maximum Efficiency, Reversible and Irreversible Processes, Carnot Cycle and Heat Engines, Clausius Inequality & Cycle Analysis
Entropy Balance: Open & Closed Systems	Entropy Generation T-S and H-S diagrams Closed System Entropy Balance Open System Entropy Balance Entropy Calculations Puzzle of Biological Order Gibbs Relations, Entropy and Heat transfer Specific Entropy Balance Equations (Liquids and Ideal Gases) Entropy Production: Integrated Systems Isentropic Processes Work and Heat Transfer Statistical Thermodynamics; Box of Air, Entropy and Poker
Thermodynamic Relations	Exact Differentials, Fundamental Thermodynamic Functions, Maxwell Relations Clayperon, Clausius-Clapeyron, Antoine Equations; Other Thermodynamic relations (J-T, sonic velocity, volume expansion, compressibility)
Power Cycles	Power Generation statistics & Energy Trends

Course Syllabus – CHE 205
Computational Methods in Chemical Engineering

1. Course number and name
   CHE 205. Computational Methods in Chemical Engineering

2. Credits and contact hours
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Ludwig C. Nitsche

4. Text book, title, author, publisher, year
   Readings and/or chapters from numerous sources are provided as PDF downloads each week on the course website. Major sources are listed below.
   a. Other supplemental materials
      Excel Scientific and Engineering Cookbook, D. M. Bourg, O'Reilly, 2006.
      Introduction to Linear Algebra, 4^{th} Ed., Wellesley Cambridge Press, 2009.
      Numerical Methods for Engineers, 6th Ed., S. Chapra & R. Canale, McGraw-Hill, 2009.
      Elementary Numerical Analysis: Algorithmic Approach, S. D. Conte & C. de Boor, McGraw-Hill, 1980.

5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Computational methods and software relevant to unit operations. Excel spreadsheets (curve fitting, heat conduction), Matlab, Aspen Plus (process simulation), algorithms and object oriented concepts in chemical engineering.
   b. Prerequisites or co-requisites
      Credit or concurrent registration in CHE 201 and MATH 210.
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Use basic Excel cell functionality to tabulate thermophysical property correlations (e.g., Shomate equation) and equations of state (e.g., Redlioch-Kwong) and numerically differentiate and integrate the results.
2	Carry out 1d (linear) and 2d (bilinear) interpolation from data tables (e.g., reduced viscosity chart, Lee-Kessler tables) by hand and with VBA macros programmed from scratch.
3	Carry out basic operations involving vectors and linear algebra (dot and cross products for vectors, matrix-vector and matric-matrix products, determinant and trace of a matrix) by hand and with Excel and MATLAB.
4	Calculate vector coordinates for changes of basis vectors by hand and with Excel.
5	Calculate and plot the Fourier series of an arbitrary function with Excel/VBA and Matlab

6	Formulate a dimensionless, 1d boundary value problem describing reaction and diffusion in a catalyst slab and calculate the Thiele modulus.
7	Solve a constant-coefficient, linear ODE using the characteristic polynomial.
8	Solve ODEs and coupled systems of ODEs With Euler's method and midpoint method.
9	Solve single and coupled nonlinear, algebraic equations using the Goal Seek and Solver functionalities of Excel.
10	Carry out error analysis of data.
11	Solve square and rectangular matrix equations arising in engineering problems with Excel and MATLAB. (Fit data by linear least squares; solve ODEs with least-squares collocation and weighted or constrained boundary conditions).
11	List (by memory) at least 5 provisions in the AIChE code of ethics.

b. Student Outcomes covered by course and measured by Performance Indicators.

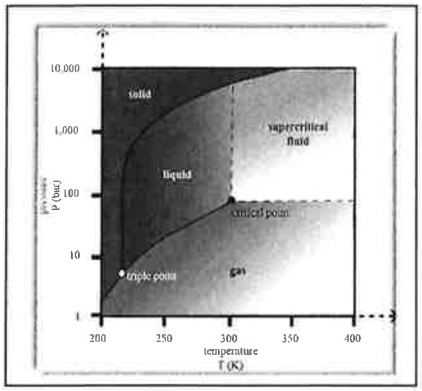
SO	PI	Description	Level
1	1-5, 8-10	Ability to apply knowledge of mathematics, science and engineering	4
2	7	Ability to identify, formulate and solve engineering problem	3
3	1-5, 8-10, 12-14	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6		Ability to analyze and interpret data	2
13	11	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Week #1. Data correlations, numerical differentiation, numerical integration, basic operations in Excel.
Week #2. Corresponding states principle. Linear and bilinear interpolation.
Week #3. Introduce VBA and syntax. Automate Lee-Kessler interpolation.
Week #4. Introduction to linear algebra. Vectors: norm, dot product and cross product. Introduce MATLAB.
Week #5. Matrix operations. Transformation of bases.
Week #6. Linear least-squares fitting. Basis functions. MATLAB.
Week #7. Rectangular matrix equations. Fourier series and Fourier projection formula for coefficients
Week #8. Constant coefficient homogeneous ODEs (characteristic polynomial). Least-squares collocation for solving ODE boundary value problems (weighted BCs)
Week #9. Least squares collocation with boundary constraints.
Week #10. ODEs. Euler's and midpoint method. Coupled ODEs. Higher order ODEs. MATLAP ODE solvers.
Week #11. Nonoinear algebraic equations. Goal Seek and Solver functions on Excel. Shooting method for ODEs. Binary chop and VBA.
Week #12. Basic statistics. Student t distribution and confidence intervals. Error Propagation
Week #13. Introduction to Aspen Plus.
Week #14. Pipe flow and friction factor. Inferring the friction factor correlation inside Aspen plus
Week #15. Heat exchangers. HeatX model in Aspen.

UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering



College:	UIC COLLEGE OF ENGINEERING	
Department:	UIC Department of Chemical Engineering	
Course:	CHE 301 Chemical Engineering Thermodynamics	
Time and Place:	CEB 230 MWF: 09:00 A.M. – 09:50 A.M.	
Instructor:	Dr. Michael Caracotsios	
E-mail:	mcaracot@uic.edu	
Cell Phone:	(630) 696-0934	
Work Phone:	(312) 413-3777	
Office Hours:	TBA	
Teaching Assistant:	Sheldon Cotts	Daniel Christiansen
E-mail:	scotts2@uic.edu	dchris28@uic.edu
Office Hours:	Th 01:00-3:00	Tu 11:00-01:00
Office Location:	CEB 130	CEB 230C

COURSE OBJECTIVES: This course is designed to extend the student's knowledge and understanding of the first and second law of thermodynamics to the critical areas of Chemical Engineering Thermodynamics. These areas encompass properties of pure fluids and fluids in homogeneous solutions, single and multicomponent phase equilibrium, and chemical equilibrium. The knowledge of this material will provide a strong foundation for all subsequent courses in chemical engineering and contribute towards understanding of chemical processes. By the end of the course the students should be able to:

- Understand the basic concepts of Internal Energy, Gibbs Energy, Entropy and Enthalpy.
- Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.
- Understand the theory of thermodynamics of ideal and non-ideal solutions
- Understand the applicability and use of various equations of state in describing the thermodynamic state of pure fluids and mixtures thereof.
- Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.
- Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

COURSE PREREQUISITES: The following are key requirements for a successful completion of this course.

- **Integral and Differential Calculus.**
- **Familiarity with a Programming Language.**
- **CHE Department Prerequisites: CHE 201 and CHE 205.**

SOFTWARE: Students may download [FREE of charge] the software package **Athena Visual Studio** from www.AthenaVisual.com. Students will also use EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

RECOMMENDED TEXTBOOKS: **|*Textbooks are Optional. My Personal Notes will be distributed to All Students*|:**

1. J. M. Smith, H. C. Van Ness and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics,* 7th Edition, McGraw-Hill (2005).
2. T. Matsoukas, *Fundamentals of Chemical Engineering Thermodynamics*, 2nd Edition, Prentice Hall (2013)
3. S. I. Sandler, *Chemical and Engineering Thermodynamics*, 3rd Edition, John Wiley & Sons, Inc. (1999)

Course Outline: CHE 301 Chemical Engineering Thermodynamics.

1. Review and Background

- Introduction to Thermodynamics in Chemical Engineering
- Elements of Differential and Integral Calculus
- Numerical Methods for Algebraic and Differential Equations
- Mass and Energy Balances in the Context of Thermodynamics
- Tutorials on Programming with VBA and Athena Visual Studio

2. Thermodynamic Properties of Single-component Fluids

- The PVT Behavior of Single-component Fluids
- The Gibbs Phase Rule for Single-component Fluids
- The Ideal Gas Behavior
- Cubic and Virial Equations of State
- Fugacity and Fugacity Coefficients from Equations of State
- Thermodynamic Properties of Single-Phase and Two-Phase Systems

3. Thermodynamic Properties of Multi-component Mixtures

- Partial Molar Properties
- Excess and Residual Thermal Properties
- The Gibbs Phase Rule for Multi-component Mixtures
- The PVT Behavior of Multi-component Mixtures
- Cubic Equations of State and Mixing Rules
- Fugacity and Fugacity Coefficients from Equations of State
- Liquid Activity Coefficients for Non-ideal Solutions
- Thermodynamic Properties of Single-Phase and Two-Phase Systems

4. Multicomponent Phase Equilibrium

- Criteria for Phase Equilibrium
- Vapor-Liquid Equilibrium: The Fundamental Problem
- Flash Calculations: PT, PH and PV Calculations
- VLE Data Analysis: Activity Coefficients from Experimental Data

5. Chemical-Reaction Equilibria

- Criteria for Equilibrium in Chemical Reactions
- Gibbs Free Energy and Equilibrium Constants
- Evaluation of Equilibrium Constants from Thermodynamic Data
- Calculation of Equilibrium Conversion for Single and Multi-reaction Systems

ACADEMIC INTEGRITY:

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community–students, staff, faculty, and administrators–share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy:
https://dos.uic.edu/conductforstudents.shtml

RELIGIOUS HOLIDAYS:

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

The University Holidays and the Religious Observances calendar can be found online at the following website: http://oae.uic.edu/religious-calendar/. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY:

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Partial credit can also be requested in cases where there is an obvious (indisputable) error made by the Professor or the Teaching Assistant.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class:

Not allowed unless explicitly stated
Repeat Offenders will be asked to leave the classroom

Cheating/Copying: The entire class gets penalized after the first warning

Homework contribution to the final grade is substantially reduced
Perpetrators will receive zero points in their homework

Late Homework and Disputes:

Late Homework will not be accepted without prior arrangements with your Professor in person
If there are Homework and Exam score disputes they will be resolved only with the Professor.

Course Syllabus - CHE 311
Transport Phenomena I

1. Course number and name
   CHE 311. Transport Phenomena I

2. Credits and contact hours
   Credit hours: 3.
   Contract Hours: Lectures: MWF, 10:00 - 10:50 am in CEB RM218; Instructor Office Hours: MWF 11:00am – 12:00pm CEB RM211; TA Office Hours: TTh 2-3pm CEB RM228

3. Instructor's or course coordinator's name
   Ying Liu, Office: CEB RM211, Telephone: 312-996-8249, Email: liuying@uic.edu

4. Text book, title, author, publisher, year
   R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2ed Ed., Wiley (2002)

5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.
   b. Prerequisites or co-requisites
      CHE 201; and credit or concurrent registration in CHE 210
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Use the mathematical language to calculate scalar, vector, and tensor operations.
2	Understand the essential physical principles of momentum, heat, and mass transport and definition of viscosity, heat conductivity, and binary diffusivity.
3	Set up differential momentum balance and analyze boundary conditions.
4	Solve constant-coefficient ODEs and apply boundary conditions to determine steady state one-dimensional flow velocity and stress profiles.
5	Solve PDEs by using the methods of separation of variables and combination of variables to result the velocity profiles of the flow flied which has two independent variables.
6	Formulate and solve the mathematical models to show how viscosity of a Newtonian fluid can be measured by various types of viscometers. Analyze the advantage and disadvantage of each viscometer.
7	Apply time-smoothed equations of change for turbulent flow.
8	Analyze the time-smoothed velocity profile near a wall.
9	Understand the physical meaning and mathematical expression of the empirical expressions for the turbulent momentum flux.

10	Understand the definition of friction factor. Use Moody diagram to solve various pipe-design problems.
11	Set up macroscopic mass, momentum, and mechanical energy balances for isothermal flow systems.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	2-8, 11	Ability to apply knowledge of mathematics, science and engineering	4
2	2, 3, 6, 9, 10	Ability to identify, formulate and solve engineering problems	4

7. Brief list of topics to be covered

Week #1: Introduction and mathematical preliminaries
Scalars, vectors, and tensors—operations involving them
Week #2: Transport coefficients (Chap. 1, 9, 17)
Fourier's law of heat conduction
Fick's law of binary diffusivity
Newton's law of viscosity
Week #3: Shell Momentum balances and velocity distribution in laminar flow
Week #4: Apply shell momentum balances on more representative laminar flows
Week #5: Conservation equations
Differential equations for mass and linear momentum
Week #6: Apply differential mass and linear momentum balance on representative flow problems
Week #7: Conservation equations for angular momentum* and energy balance
Week #8: Macroscopic balances for mass and momentum
Week #9: Macroscopic balances for mechanical energy
Week #10: Velocity distributions with more than one independent variable
Time dependence
Stream function
Week #11: Similarity analysis
Week #12: Introduction of turbulent flow
Week #13: Time-smoothed velocity profile for turbulent flow near a wall
Week #14: Friction factors for flow in a circular tube
Week #15: Friction factors for flow around a sphere

CHE 312: Transport Phenomena II: Heat and Mass

Course Syllabus – Fall 2018

Class meetings:	Monday, Wednesday, Friday 1:00 pm - 1:50 pm	CEB 214

Instructor:

Brian P. Chaplin, Ph.D.: 221 CEB, (312) 996-0288 (office), chaplin@uic.edu
Office hours: TBA; or by appointment

Teaching Assistant:

Soroush Almassi: 115 CEB, (312) 996-0330 (office), salmas3@uic.edu
Office hours: TBA; or by appointment

Objectives and Prerequisites

Introduce students to heat and mass transport phenomena. Specific topics include heat conduction, convection, and radiation, heat exchanger design, diffusion, and mass transfer coefficients.

Prerequisite: CHE 311, Differential Equations

Required Text

Bird, R.B., Stewart, W.E., and Lightfoot, E.N., 2007. *Transport Phenomena*, Revised 2nd Edition, Wiley and Sons. ISBN-13: 978-0470115398

Grading policy

The course grade will be based on the following:

In-term exams (3)—45%
Assignments and Pop Quizzes—20%
Final Exam—35% (Monday, Dec. 10th, 1:00p-3:00p)

Attendance policy

1. Attendance at class meetings is highly recommended.
2. If you miss a class meeting, you are responsible for the course material covered and compliance with any announcements made. Most handouts will be posted online.
3. Attendance at examination periods is mandatory. If you miss a quiz or examination, you will receive a zero for that quiz/examination. Makeup quizzes/exams will **ONLY** be given, in the case of an extreme emergency. I will make all decisions on this matter on a case-by-case basis.

Assigned Work

In this course, all assigned work, unless otherwise stated, should be completed and handed in individually. Evidence of direct copying will be addressed in accordance with the University's Academic Integrity policies (see below). Required assignments will be due at 4 pm on the due date. You are asked to prepare all work in a professional manner. Late work assignments will **NOT** be accepted.

Solutions for homework problems will be available for student review, and will be posted on Black Board after the assignment due date. Students are encouraged to **first** compare homework and the correct solution to find the source of errors in their calculations. If there is still some confusion, the student is encouraged to meet with the TA or myself. Office hours provide an ideal opportunity to discuss such matters.

Re-grade Policy: There will not be any exceptions to the following policy. If you would like an assignment (i.e., homework, quiz, or exam) re-graded, you must submit a typed explanation of your dispute along with the original graded assignment to me within one week of the *return date*. The *return date* is defined as the date that I handed the graded assignment back to students in class. This policy applies even if you were NOT in class and did not receive the graded assignment. Your entire assignment will be re-graded, and as a result your grade may increase, decrease, or stay the same.

Exam Policies

There will be three in-term exams during the semester. Any exam absences will receive a grade of **ZERO**. All exams may be cumulative, although they will emphasize the material covered since the last exam.

All cell phones must be stowed away from your chair/desk/person. The following actions are prohibited during a quiz or exam. (1) Using a cell phone (or PDA, etc.) in any manner or having it in your hand or on/under your desk. (2) Opening a browser window and/or accessing the Internet. (3) Talking with another student or passing notes.

If a student misses the final exam, they will receive a grade of **ZERO**. A make-up final exam will not be given. The student feedback assessment survey, to be conducted during the 15th week of the semester, will count for 3 points (out of 100 points in total) on the final exam.

Academic Integrity

This course and its associated coursework are being administered under the policies of the University of Illinois at Chicago (UIC) Student Code of Conduct. In keeping with UIC policy, **evidence of academic dishonesty may result in a failing grade for the course and disciplinary review by the University**. Academic dishonesty includes, but is not limited to, cheating (giving or receiving aid), fabrication/falsification, plagiarism (including not accurately referencing source material), bribes, favors or threats, examination by proxy (taking an exam for someone else), grade tampering and submitting non-original works authored by someone other than the student. Refer to the student handbook or Student Disciplinary Procedures (www.uic.edu/depts/sja) for definitions and details. For a complete review of UIC policies regarding intellectual integrity and academic honesty, please visit: http://www.uic.edu/depts/dos/studentconduct.html.

Course Topics

Heat Transfer:

- Fourier's Law of Heat Conduction (Section 9.1)
- Thermal Conductivity (Sections 9.2, 9.5)
- Shell Energy Balances and Temperature Distributions in Solids (Sections 10.1-10.3, 10.6-10.7)
- Unsteady Heat Conduction in Solids (Section 12.1)
- Heat Transfer in Fluids (Sections 9.7-9.8, 10.4, 10.5, 10.8)
- Equations of Change for Nonisothermal Systems (Sections 11.1, 11.2, 11.4)
- Free Convection (Sections 10.9, 11.3, 11.4-5)
- Dimensional Analysis (Section 11.5)
- Temperature Distributions in Turbulent Flow (Ch. 5, Sections 13.1-13.3)
- Heat Transfer Coefficients (Sections 14.1-14.7)
- Macroscopic Balances for Nonisothermal Systems (Sections 15.1-15.4)

Mass Transfer:

- Diffusivity and Mass Transport (Sections 17.1, 20.1)
- Concentration Distributions in Solids and Laminar Flow (Sections 18.1-18.7)
- Equations of Change for Multicomponent Systems (Sections 19.1-19.5)
- Mass Transport Coefficients (Sections 22.1-22.2)
- Macroscopic Balances for Multicomponent Systems (Sections 23.1-23.2)

Important Dates

In-term Exams (Subject to Change): Oct. 3rd, Nov. 2nd, Nov. 30th
Final Exam: Monday, Dec. 10th, 1:00p-3:00p

AIChE Code of Ethics

Members of the American Institute of Chemical Engineers shall uphold and advance the integrity, honor and dignity of the engineering profession by:

- Being honest and impartial and serving with fidelity their employers, their clients, and the public;
- Striving to increase the competence and prestige of the engineering profession;
- Using their knowledge and skill for the enhancement of human welfare.

To achieve these goals, members shall:

1. Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties.
2. Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.
3. Accept responsibility for their actions, seek and heed critical review of their work and offer objective criticism of the work of others.
4. Issue statements or present information only in an objective and truthful manner.
5. Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality.
6. Treat fairly and respectfully all colleagues and co-workers, recognizing their unique contributions and capabilities.
7. Perform professional services only in areas of their competence.
8. Build their professional reputations on the merits of their services.
9. Continue their professional development throughout their careers, and provide opportunities for the professional development of those under their supervision.
10. Never tolerate harassment.
11. Conduct themselves in a fair, honorable and respectful manner.

CHE 321
Chemical Reaction Engineering
Spring 2018

INSTRUCTOR: Vikas Berry (312) 996-2342, vikasb@uic.edu

MEETING TIMES: TU 9:30 – 10:45 p.m., **ROOM:** 2LCC C004

TEXT: "Elements of Chemical Reaction Engineering", **Fifth, Fourth or Third Edition**, by Fogler, H. Scott

OFFICE HOURS: Wednesday 2:00 – 3:30 PM (Room 205 CEB)

TA: Sheldon Cotts (T: 1:30-3PM, Room 130 CEB), and Rousan Debbarma

COURSE OBJECTIVES:
This course is designed to introduce chemical engineering students with reaction engineering and to develop an expert knowledge about the various aspects of chemical reactions and kinetics. This is one of the most important courses in Chemical Engineering curriculum and will strengthen the foundation of a chemical engineering students.

Upon satisfactory completion of this course, the students will be able to:

1. Compare and contrast batch reactors, continuous stirred tank reactors (CSTR), and plug flow reactors (PFR).
2. Derive and solve the mass balance for isothermal reaction in a CSTR, PFR, or batch reactor.
3. Calculate the equilibrium conversion in any reactor type for isothermal or adiabatic conditions.
4. Set up and solve the governing mass and energy balances for a CSTR, PFR, packed bed, or batch reactor.
5. Estimate reaction orders, reaction rate constants, and Arrhenius parameters from experimental kinetic data.
6. Set up and solve the governing mass balances for multiple reactions in a CSTR, PFR, or batch reactor
7. Suggest which type of continuous reactor (CSTR or PFR) is advantageous for maximizing product selectivity for series or parallel reactions
8. Derive a Langmuir-Hinshelwood kinetic rate law
9. Incorporate a mass transfer coefficient into a reaction rate expression to account for external mass transfer limitations
10. Calculate the effectiveness factor for a catalyst particle given physical properties of the particle (pore size, tortuosity, particle diameter), the reaction rate constant, and the diffusion coefficient for the reactant
11. Suggest experiments that could be used to determine whether external and internal mass transfer is important for a heterogeneous reaction
12. Calculate the residence time distribution from a tracer study.

13. Draw similarity between electron transport through nanoparticles and chemical reactions.

GRADES:

Grades will be based on:

5 Quizzes	5 %	1 % X 5
2 Regular Exams	40 %	20 % X 2
1 Final Exam	25 %	25 % X 1
1 Project Report	12 %	12 % X 1
1 Survey Report	3 %	3% X 1
10 Homeworks	15 %	1.5 % X 10
	100 %	

QUIZZES: There will be five in-class quizzes (best 5 out of 7). **There will be no make-up quizzes.**

EXAMS: Exams will be taken during normally scheduled class periods. All the exams will be in the classroom and will consist of two parts: *Closed Book Part* and *Open Book Part.* Any necessary information relating to conversion factors and component properties will be provided for the closed book part of the exam. Test problems will be derived from lectures and homework problems. Use of only portable-hand-held calculator is permitted. The calculator must be used only for basic arithmetic calculations. No text information or graphs stored on the calculator must be used during the test. Please bring your own paper.

Each exam is worth 20% and each will cover half of the course. **There will be no make-up exams.** The final exam will be worth 25% grade and will be comprehensive.

If you feel the need for re-evaluation or contesting the score on an exam, you will have 10 days after you get your exam scores. After 10 days, your grade will be considered final.

PROJECT: Each student will work on a project. The project statement is provided at the end of this syllabus.

SURVEY: Each student needs to contact (at least) two recent graduates working in a chemical engineering company (one such person can be interviewed only by a maximum of two students) and interview them about their experience with finding a chemical engineering job and about the nature of their job. Then make a report on (a) what you learnt (about finding ChE job and excelling in a ChE job), and (b) make a plan on how you will find a job (or pursue higher education). You need to include a signed letter of confirmation of discussion from the interviewees.

HOMEWORKS: There will be a total of 10 homework assignments of 1.5 points each (best 10 out of 11).

Course Timetable

Lecture	Date	Lecture Title / Activity	Submissions/ HW/Q Dates
1	16-Jan	Course Introduction, Reaction fundamentals	
2	18-Jan	Reactor Types, General Mole Balances, Conversion	
3	23-Jan	Conversion, Reactor Sizing, Stoichiometric Tables	
4	25-Jan	Stoichiometric Tables	Q1
5	30-Jan	Isothermal Reactor Design	HW 1
6	1-Feb	SOFTWARE - Polymath and MaTLAB for Reactor Design	Q2
7	6-Feb	Pressure Drop in Reactors	HW2
8	8-Feb	Unsteady State Operation, Rate Analysis	Q3
9	13-Feb	Multiple Reactions, I	HW3
10	15-Feb	Multiple Reactions, II	
11	20-Feb	General Energy Balance & CSTR	HW 4
12	22-Feb	Review for Exam 1	
	27-Feb	EXAM 1	
13	1-Mar	PFR Energy Balance , Batch Energy Balance	
14	6-Mar	Equilibrium Conversion	HW5
15	8-Mar	Non-Isothermal Reactor Example	Q4
16	13-Mar	Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions	
17	15-Mar	Summary of Non-Isothermal Reactors	HW6
18	20-Mar	Catalytic Reactions, Langmuir-Hinshelwood Rate Laws	Q5
19	22-Mar	Langmuir-Hinshelwood Rate Laws - II	
	Break		
20	3-Apr	Heterogeneous Reactors, Mass Transfer	HW 7
21	5-Apr	External and Internal Mass Transfer Limitations	HW8
22	10-Apr	Effectiveness Factor and Falsified Kinetics	Q6
23	12-Apr	TA - Review of Exam 2	HW9
	17-Apr	EXAM 2	
24	19-Apr	TA - Residence Time Distribution II	
25	24-Apr	Zero-Parameter Flow Models	
26	26-Apr	Non-Ideal Flow	HW10, Q7
27	1-May	Combination of Ideal Reactors	HW11
28	3-May	Review of Exams	Project Due
		Final Exam	

Homework # 1

1. A CSTR reactor has to be designed for the following third-order, liquid-phase reaction:
   $3A + B \rightarrow C + D$ (1^{st} order with respect to A and 2^{nd} order with respect to B, $k = 0.2\ L^2/mol^2/s$)
   Stoichiometric feed of A and B enters a reactor along with an inert I. The total inlet molar flow rate of 3000 mol/min. The mole fraction of A in the inlet, $x_{AO} = 0.60$, and v_O is 100 L/min.
   (a) Make the stoichiometric table for the species.
   (b) Find the CSTR volume required to achieve a conversion of 0.8.
2. A *reversible*, gas-phase reaction ($A \leftrightarrow B$) is taking place in a PFR. The inlet concentration of A is 3 moles/L, $v_o = 0.5$ L/s, with no B in the feed. The forward reaction rate constant is 3 **mol/m³/atm/s** and the equilibrium constant is 1. If the total reactor pressure is 3 Atm:
   (a) What is the maximum conversion possible in this reaction?
   (b) Find an expression for conversion Vs volume of the reactor? For what volume of the reactor will the conversion be 0.42? [Note the units of the reaction constant]

Homework # 2

1. An isothermal reversible reaction ($A \leftrightarrow B$) is carried out in an aqueous solution. The reaction is first-order in both directions. The forward rate constant is 1 h^{-1} and the equilibrium constant is 1. The feed to the plant contains 100 kg-mol/m³ of A and enters at the rate of 120 m³/h. The reactor is a stirred tank of volume 180 m³. The exit stream then goes to a separator, where one stream of 40 m³/h with double the concentration of A at the exit of the reactor (C_A) is sent back to the CSTR in a continuous operation. The rest of the stream from the separator becomes the system exit (no reaction occurs in the separator).
   a. Find the system conversion.
   b. Compare this conversion with what you will get if there is no recycle.

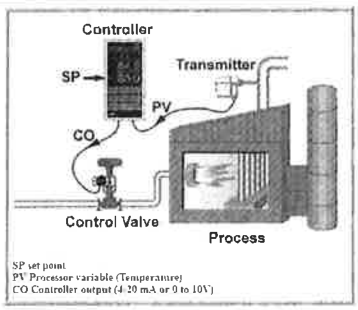
Homework # 3

1. For a first order reaction ($A \rightarrow B$) in a PFR ($k = 1\ s^{-1}$, $\nu = 10\ dm^3/s$, $V = 100\ dm^3$), a fraction f of the exit volumetric flow rate ν is fed back to the PFR. Plot f as a function of conversion (χ).
2. (40 points) The following reactions take place in a batch reactor.
   $A \leftrightarrow D$, $\quad -r_{1A}=k_1[C_A - C_D/K_{1A}]$
   $A \leftrightarrow U$, $\quad -r_{2A}=k_2[C_A - C_U/K_{2A}]$
   $k_1 = 1.0\ min^{-1}$, $K_{1A} = 5$
   $k_2 = 100\ min^{-1}$, $K_{2A} = 2$
   $C_{A0} = 2\ mol/dm^3$

   (a) Plot and analyze conversion and the concentrations of A, D, and U as a function of time. When would you stop the reaction to maximize the concentration of D?
   (b) When does the maximum concentration of U occur?
   (c) What are the equilibrium concentrations of A, D, and U?
   (d) What would be the exit concentrations from a CSTR with a space time of 1.0 min? of 10.0 min? of 100 min?

UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering



College:	UIC COLLEGE OF ENGINEERING	
Department:	UIC Department of Chemical Engineering	
Course:	**CHE 341 Chemical Process Control**	
Time and Place:	CEB 230 MWF: 12:00 P.M. – 12:50 P.M.	
Instructor:	**Dr. Michael Caracotsios**	
E-mail:	mcaracot@uic.edu	
Cell Phone:	**(630) 696-0934**	
Work Phone:	**(312) 413-3777**	
Office Hours:	CEB 209 MWF: 02:00 P.M. – 04:30 P.M.	
Teaching Assistants:	**Mina Shahmohammadi**	**Azadeh Amiri**
E-mail:	mshahm2@uic.edu	aamiri2@uic.edu
Office Hours:	**Th: 02:00 PM – 04:00 PM**	**Tu: 10:00 AM – 12:00 PM**
Office Location:	**Lunch room**	**CEB 121**

COURSE OBJECTIVES: This course is designed to extend the student's understanding of the steady-state macroscopic and microscopic material and energy balances into the critical area of process dynamics. The major objective of the course is to emphasize and address the following issues pertinent to the safe and cost effective operation of chemical plants. Given the process and operational targets, the course objectives would be to design a control system which is (a) stable, (b) has good performance characteristics, and (c) is robust. These design attributes require knowledge of dynamic behavior of processes (process modeling, solution of dynamic equations, characterization of dynamic behavior); control systems; stability and techniques for assessing it; performance criteria and how they are affected by controller parameters. By the end of the course the students should be able to:

- Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- Know how to develop empirical models from process data using the step response methodology
- Understand the theory of feedback and feed forward control and determine optimal controller settings for stable and robust performance.
- Know when to select and how to implement cascade and feed forward controllers
- Apply available computer software (Matlab, Excel or Athena Visual Studio), or use Laplace transformation theory to solve process dynamic models and investigate their open and closed loop response to various external disturbances.

COURSE PREREQUISITES: The following are key requirements for a successful completion of this course.

- **Integral and Differential Calculus Courses.**
- **Linear First and Second Order Differential Equations.**
- **Familiarity with a Programming Language.**
- **CHE Department Prerequisite(s): MATH 220, CHE 312, CHE 313 and CHE 321.**

SOFTWARE: Students may download [FREE of charge] the software package **Athena Visual Studio** from www.AthenaVisual.com. Students may also use EXCEL or MATLAB.

RECOMMENDED TEXTBOOKS: |*Textbooks are Optional. My Personal Notes will be distributed to All Students*|:

1. D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, *Process Dynamics and Control*, 3rd Edition, Wiley, 2011.
2. T.E. Marlin, *Process Control: Designing Processes and Control Systems for Dynamic Performance*, McGraw-Hill, 2000.
3. B.A. Ogunnaike and W.H. Ray, *Process Dynamics, Modeling and Control*, Oxford, 1994.
4. C.A. Smith and A. B. Corripio, *Principles and Practice of Automatic Control*, Wiley, 1985.
5. G. Stephanopoulos, *Chemical Process Control: An Introduction to Theory and Practice*, Prentice Hall, 1984.

COURSE OUTLINE: CHE 341 CHEMICAL PROCESS CONTROL.

1. Math & Programming Review. Material and Energy Balances.

- Introduction to Process Control in Chemical Engineering
- Elements of Differential and Integral Calculus
- Numerical Methods for Algebraic and Differential Equations
- Mass and Energy Balances in the Context of Process Control
- Tutorials on Programming with VBA and Athena Visual Studio

2. Control Concepts and Dynamic Modeling

- Development of Dynamic Process Models
- Solution of Dynamic Models and Open Loop Response
- Laplace Transforms and Transfer Functions
- Linearization of Process Dynamic Models
- Process Identification from Plant Data

3. Feedback Control Fundamentals

- P, PI and PID Controllers
- Block Diagrams and Routh-Hurwitz Stability Criterion
- Instrumentation Hardware and Representation

4. Feedback Controller Design and Tuning

- Criteria for Controller Tuning
- PID Controller Tuning Methods
- Guidelines and Troubleshooting Common Control Loops

5. Advanced Control Systems and Multi-Input Multi-Output Systems

- Feedforward and Cascade Control
- Interaction and Loop Pairing and Decoupling – Relative Gain Array
- Control of Common Industrial Process Units
- Model Predictive Control of Linear Processes
- Advanced Process Control and Real-Time Optimization

ACADEMIC INTEGRITY:
As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community-students, staff, faculty, and administrators-share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy:
http://dos.uic.edu/docs/Student%20Disciplinary%20Policy.pdf

RELIGIOUS HOLIDAYS:
Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

The University Holidays and the Religious Observances calendar can be found online at the following website: http://oae.uic.edu/religious-calendar/. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY:
Partial credit can be requested under certain circumstances. The student should make an appointment with the TA/Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Partial credit can also be requested in cases where there is an obvious (indisputable) error made by the Professor or the Teaching Assistant.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY
Talking/Smart Phones/Laptops/Food and Drinks in Class:
Not allowed unless explicitly stated
Repeat Offenders will be asked to leave the classroom
Cheating/Copying: The entire class gets penalized after the first warning
Reduce the homework contribution to the final grade
Perpetrators will receive zero points in their homework
Late Homework and Disputes:
Late Homework will not be accepted without prior arrangements with your Professor in person
If there are Homework and Exam score disputes they will be resolved only with the TA/Professor.

CHE 381/382

Course Description: The laboratory component of the Chemical Engineering curriculum. CHE381 and CHE382, also known as Unit Operations Laboratory I and II, provides students with an exceptional hands-on opportunity to work with physical equipment in a laboratory environment. Students, organized into small groups, conduct experiments, write pre-lab plan and final lab reports, and present their work to their colleagues.

Text: Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott 7th Edition, McGraw-Hill, 2005.
Optional: Writing Styles and Standards in Undergraduate Reports, J. Donnell, S. Jeter, C. MacDougall, J. Snedeker, 3rd Ed. College Publishing, 2016.

Aims, Scope, Level

Provide students with a hands-on opportunity to work with chemical engineering process equipment in a laboratory environment. Students will learn to solve complex, open-ended engineering problems in a team setting, integrate knowledge and skills from prior chemical engineering courses, design and conduct realistic experiments, analyze data, draw concrete conclusions, make relevant recommendations, communicate findings effectively through the written and oral word, and apply safety practices in a laboratory environment.

Senior-level course; Students must have the necessary background in mass and energy balances, mass transfer, heat transfer, momentum transfer, separation processes, chemical engineering thermodynamics and analytical chemistry.

Grading:

20% Lab Prep Report- 4 page reports including Safety Evaluation
40% Final Lab Report: 10 page report including Error Analysis
10% Oral Presentation: Every two weeks given to next lab team; final presentation to whole class end of semester
10% Safety Quiz; 20% Individual Performance Assessment

Assessment of written reports will be based on the following attributes:

- Completeness (inclusion of all relevant content and report sections specified)
- Organization (orderly and logical flow of material), clarity, conciseness including report length, and grammatical correctness.
- Technical correctness of calculations
- Technical correctness of analysis, results, and conclusions
- Initiative (original work to more efficiently achieve objectives or go beyond basic objectives).

Assessment of oral reports will be based on the following attributes:

- Overall presentation quality
- Accuracy and clarity of statements
- Quality of data presentation and analysis
- Degree of technical comprehension

Assessment of Individual Performance is based on:

- Attendance,
- Individual team member contributions to each experiment,
- Team member assessment

List of Experiments

CHE381	CHE382
1. Viscosity: Non-newtonian fluids	1. Distillation
2. Evaporation	2. Ultrafiltration
3. Fluid Flow in Pipe Systems	3. Process Control
4. Fluidized Bed	4. Gas Absorption
5. Heat Transfer	5. Liquid-Liquid Extraction
6. Membrane Separation	6. Humidification

ASSIGNMENTS AND LAB EXPERIMENTS

Industrial Chemical Hygiene Plan	CHP from an industrial R&D facility. Provide additional safety learning and comprehension. 20 question quiz
Guidelines on Hazardous Chemical Safety	Select chapters from JT Baker Hazardous Chemical Safety course; toxic, flammable, corrosive, and reactive chemicals, insidious hazards, and compressed gases. 20 question quiz
SACHE Tutorials	Dust Explosion tutorial Chemical Reactivity tutorial 50 question quizzes
	LAB EXPERIMENTS and TOPICS COVERED
Viscosity	Capillary tube viscometer, Non-newtonian fluid, Ostwald-DeWaele equation, apparent viscosity
Fluid Flow in Pipe Systems	Pipe Demonstrator; Pressure drop in straight pipes, orifice meter, pipe fittings; laminar/turbulent flow, Bernoulli equation, friction factor, loss coefficients, Moody Chart, Colebrook equation, rotameter calibration
Evaporation	Single-effect evaporator, 10 wt% sugar/water, solute/solvent separation; mass and energy balances, effect of vacuum pressure on boiling point, capacity and economy
Fluidized Beds	Solid-air fluidized bed; fluidization behavior of sand and ion-exchange resin beads; bed height, pressure drop, Ergun equation, particle size distribution
Heat Transfer	Steam-jacketed steel-walled constant-stirred tank; conductive and convective heat transfer, unsteady state heat transfer, baffled and unbaffled systems
Membrane Air Separation	Polysulfone membrane module system; N_2/O_2 separation; solution-diffusion phenomena, permeability/mass transfer coefficients, recovery coefficient
Distillation	8-stage column, 10 wt% ethanol distillation; total reflux and reflux ratios, boil-up rate, vapor-liquid equilibrium, McCabe-Thiele diagram, column efficiency, refractometer measurements and calibration.
Ultrafiltration	Single-column unit; milk/water separation, pressure effects, permeability, concentration polarization, efficiency
Gas Absorption	Gas-liquid packed bed absorption column; CO_2/ Air/Water system; Packed bed pressure drop, flooding characteristics; CO_2 absorption/desorption, mass transfer cofficients, titration measurements, Hempl apparatus
Liquid-Liquid Extraction	3-stage extraction; acetic acid in mineral oil/water system; extraction efficiency, McCabe-Thiele analysis, ternary liquid equilibrium diagrams, titration measurements
Humidification	Falling-water film humidification column; water/air system; temperature and flow rate effects on humidification, combined heat/mass transfer, Sherwood number correlation, psychrometric chart.
Process Control: Tank Level	Level control process unit; water level control in a tank; process control theory and practice; Setpoint and disturbances; manual control (open loop), proportional and proportional-integral control (closed-loop),

UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING

CHE 396 – SENIOR DESIGN I
Fall 2018

Lecture:
Section 1: T/Th: 10:00am – 11:50am: 230 CEB
Section 2: T/Th: 1pm-2:50pm: 230 CEB

No Classes:
November 22nd (Thanksgiving Break)

Office Hours:
12pm-1pm Tuesdays and Thursdays by appointment (Dr. Bilgin)
By appointment with TA

Instructor:
Dr. Betul Bilgin, bbilgin@uic.edu

Teach Asst:
Chenxian Xu, cxu41@uic.edu

Prerequisites:
CHE 312, CHE 313 and CHE 312

Text:
Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler) (required)

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009 (as a reference)

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker (printed book ISBN: 9781932780093 / retail price: $39 or e-book ISBN: 9781932780109 / retail price $19) (as a reference)

Course Description:
This is an introduction class to modern process design and development, engineering economics and report writing. It also covers design and cost of equipment related to materials handling to heat transfer and reactors.

Course Objectives:

- Provide a real-world perspective on process design
- Introduce design and selection of equipment
- Give insight into practical constraints that govern design decisions
- Help you translate knowledge into practice & start solving real problems
- Improve presentation, analytical thinking and team working skills

Course Outline:

Date	**Chapter**	**Assignment**
August 28th 2018	Syllabus Development	
August 30th 2018	Chapter 1 (Introduction)	
September 4th 2018	Chapter 1 (Introduction)	
September 6th 2018	Chapter 2 (Flowsheet development)	1st Assignment will be handed
September 11th 2018	Chapter 2 (Flowsheet development)	
September 13th 2018	Chapter 2 (Flowsheet development)	
September 18th 2018	Chapter 3 (Utilities and Energy Recovery)	
September 20th 2018	Chapter 3 (Utilities and Energy Recovery)	1st Assignment Due 2nd Assignment will be handed
September 25th 2018	Chapter 4 (Process simulation)	Presentation 1
September 27th 2018	Chapter 4 (Process simulation)	Presentation 1
October 2nd 2018	Chapter 5 (Process control)	Quiz 1
October 4th 2018	Chapter 5 (Process control)	2nd Assignment Due
October 9th 2018	Chapter 6 (Materials of construction)	Quiz 2 3rd Assignment will be handed
October 11th 2018	Chapter 6 (Materials of construction)	
October 16th 2018	Chapter 7 (Capital cost)	Presentation 2 will be started
October 18th 2018	Chapter 7 (Capital cost)	3rd Assignment Due 4th Assignment will be handed

October 23rd 2018	Chapter 8 (Operating cost)	Quiz 3
October 25th 2018	Chapter 8 (Operating cost)	
October 30th 2018	Chapter 9 (Economic Analysis)	
November 1st 2018	Chapter 9 (Economic Analysis)	4th Assignment Due 5th Assignment (about ethics) will be handed
November 6th 2018	Chapter 9 (Economic Analysis)	Quiz 4
November 8th 2018	Chapter 10 (Process Hazard, Professional ethics)	
November 13th 2018	Chapter 10 (Process Hazard, Professional ethics)	
November 15th 2018	Chapter 10 (Process Hazard, Professional ethics)	5th Assignment Due
November 20th 2018	Professional Ethics Presentations (Quiz 5)	6th Assignment will be handed
November 22nd 2018	NO Class	
November 27th 2018	Chapter 11 (General Site Consideration)	
November 29th 2018	Chapter 12 (Optimization)	6th Assignment Due Quiz 6
December 4th 2018	Review	
December 6th 2018	Review	Final Project Due

Class Structure:

In a typical class,

-You are required to read the topic before the class
-During half of class time, I will present key points of the topic
-After presentation of the topic, I will give you a small project design that you will be working in the class with a group of 5.
- After 15-20 minutes of group discussions, one of the groups will be randomly selected and they will present their design.

In-class projects are expected to be the product of team efforts. Each member of the team is expected to contribute equally to the development of the design. In- class projects will not be graded. Feedback will be provided by the instructor.

Homework / mini-project assignments:

There will be 6 biweekly assignments in fall semester. For each assignment, you will be working with a group of total four students who will be randomly selected. Therefore, you will have the opportunity to work with different groups of people and to practice team work. I expect that all the members of the group contribute equally to the group's project. Homework due dates are listed on the syllabus. They are due until the beginning of the class on due date. Late homework will be accepted up to 24 hours after due date for 50% credit. Solutions to homework will be presented in the class by students. Instructor will be randomly selecting which group will present which homework question. You should be prepared to present any question of homework. Also, peer evaluations will be very important. Your peers in particular homework will be evaluating your performance. Those evaluations will be graded and it will affect your homework grade. Peer evaluations will be done on catme.org and average evaluation scores of group members will be calculated. Each group members' grade will be calculated by multiplying group grade by peer evaluation average percentage of the individual.

Final Project/ Proposals (written):

A major design project will be assigned to each group of students at the end of October. Student groups will include 4 members. Each group will have their own design project. Each group is required to submit a proposal at the end of semester. Proposal should include background material about the project, explanation of your design objective and process description. Written proposal will be due the end of semester.

Exams:

There will be NO exam for this course.

Presentations:

Students are required to give two oral presentations throughout the course. First presentation will be based on HW 1 topic for 20 min. Second presentation will be group of three for 15 min. Topic choices for second presentation will be posted. Contents should be history, production, and design of the particular product. Presentation should be prepared for general public audience as well as for chemical engineers. Presentations will be graded based on the design of the presentation, how you form the story and how you tell the story; it will not only be graded based on the content. Presentations will be graded by the instructor. Classmates will be providing feedback.

Quizzes :

There will be 6 individual quizzes throughout the course. Quizzes will be based on HW and presentations. Quizzes are for assessing individual learning.

Attendance/Participation:

Students are expected to attend all classes, and attendance is incorporated as 10% of the final grade. Students are allowed two excused absences or one unexcused absence. If absences exceed these limits, the student will receive a zero for attendance. Participation to class is also important. It will be incorporated to attendance grade. In-class assignments will be graded as participation.

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for ChE 396:

-Inherently safer design
-Safety in the process industries

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance/ Participation: ....10% (in-class assignments will be incorporated to participation)
Homework/Mini-Project Assignments:...........45% (6 of them, group of 4, written; individual grade will be calculated based on multiplication of group grade with peer evaluation percentage)
Oral Presentations: ....20% (2 times, first one is based on HW 1 so group of 4, second one is group of 3)
Quizzes: ....15% (6 of them, individual, based on HWs)
Final Project/ Proposal:........................... 10% (written only)

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Most relevant student outcomes for this course:

SO	Description
1	Ability to apply knowledge of mathematics, science and engineering
2	Ability to identify, formulate and solve engineering problems
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice

7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
9	Knowledge of contemporary issues
10	Recognition of the need for and ability to engage in life-long learning
11	Ability to function on multidisciplinary teams
12	Ability to communicate effectively
13	Understanding of professional and ethical responsibilities

UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING

CHE 397 – SENIOR DESIGN II
Spring 2018

Lecture:
Section 1: T/Th: 11am – 12:15am: 230 CEB
Section 2: T/Th: 1pm-2:15pm: 230 CEB

No Classes:
March 27th and 29th (Spring break)

Office Hours:
Tuesdays and Thursdays by appointment (Dr. Bilgin)

Instructor:
Dr. Betul Bilgin, bbilgin@uic.edu

Teach Asst:
Azadeh Amiri,

Text:
Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler)

Course Description:
Application of principles and design methodology of chemical engineering to the design of large-scale chemical processes and plants. A major design project is assigned for solution and presentation by students working in small groups.

Requirements/Constraints:
-All the calculations need to be submitted in excel
-All the work need to be archived in blackboard and all the mentors will have access to them
-There will be 4 reports through the semester.
-There will be 4 presentations through the semester. During presentations, all the group members are expected to participate.
- Final report needs to be submitted to the mentors 2 weeks prior to the end of semester.
- 7 stepping stones for the project
    -1. Process description
    -2. Material/energy balances and PFDs
    -3. More detailed equipment calculations (at this point, student are encouraged to contact with vendors)
    -4. Pricing equipment, economics, cash flows

-5. Control scheme
-6. Safety/potential hazardous materials list
-7. Final report

Report/Presentation Deliverables:

Report 1:

- Introduction (your goal is to get the reader excited to read your report)
  - Clear definition of problem statement
  - Why this problem is important?
  - Explanation of your goals and objectives
  - Market analysis
  - Clear explanation of judging criteria (more environmental friendly, more economical etc.)
- Process description (your goal is make the reader believe that you already did the research on possible ways to make the product, you analyzed them well and have chosen your way critically)
  - What is the problem?
  - What are possible solutions? (competing processes (strengths/weaknesses of them), chemistries) (graphs, figures, tables would be helpful for comparison)
  - What is your solution?
  - Why you have chosen that solution?
- Appendix 1(design basis)
- Appendix 2 (BFD and material balance)
- Preliminary appendix 3 (PFD) (For report 1, it is only preliminary PFD)
- Preliminary appendix 4 (material balances) (For report 1, it is only rough material balance meaning that mass balance block in and out. Estimates of products, by-products, waste etc.)

Report 2:

- Final PFD
- Detailed mass balance (This means detailed inputs and outputs to each part of the process along with conditions ie, temperature, pressure, etc.)
- Equipment list (Most of the major equipment should be identified along with a good start on the equipment list)
- Equipment sizing (column sizes, pump and compressor sizing, heat exchanger sizing, along with horsepower requirements and heating/cooling loads) (One distillation tower sizing or separation unit sizing should be done by hand calculation. Hand calculation and Aspen results should be compared and analyzed critically)
- Preliminary utilities demand summary for entire plant.

Report 3:

-Process controls
  -Describe control objectives
  - What are the main process parameters?
  - What is your control strategy?
- Environmental and safety concerns
  -List environmental issues, air emissions, water discharges, etc.
  -What design features you would use to mitigate the effects of these discharges?

-Outline the process hazards which you must legally and ethically address. (That means, for instance, all OSHA PSM highly hazardous materials should be listed, including quantities on site. Process hazards associated with toxic, explosive, hot and cold process streams should be identified and a safeguard included in the design.)

-Cost analysis

-detailed cost analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen econ analysis, etc..) Analyze your results rather than just listing the numbers.

Report 4 (Final Report):

-Corrections/updates of report 1,2 and 3.
-Utilities
-Executive summary
-Recommendations
-Heat exchanger network(optional)

Assigning groups

Every group will include 4-5 members.
Students will choose one person who they want to work with and one who they do not want to work with.
Dr. Bilgin will assign the groups based on student's choices.
Each group will be assigned to one mentor.

Responsibilities

-Each group is responsible to meet with their mentor weekly (physical meeting or online meeting or phone conversation).
-Each group is responsible to choose a group leader and rotate the leader among the members of the group every three weeks. Leader's responsibility is to set up the meetings with mentor.
-Before the meeting; each group is required to submit a meeting agenda to corresponding mentor 2 days before the meeting. If the agenda is not submitted, then meeting will be canceled.
-After the meeting; each group is required to submit meeting notes to corresponding mentor next day after the meeting.
-Presentations will be 15 min. There are 9 groups in morning section and 9 groups in the afternoon sections. Morning group presentations will be on Tuesday/ Thursday 10am-12.30pm. Afternoon group presentations will be on Tuesday/ Thursday 1-3.30pm as announced on the course blackboard site.
-Intermediate reports will be reviewed by professor.
-Final reports will be reviewed by assigned mentor of each group.

Course Outline:

Date	Concept	Assignment
January 16th 2018	Syllabus review	
January 18th 2018	UIC Career Center	Guest speaker
January 23rd 2018	Report Format by Jerry Palmer	Guest speaker
January 25th 2018	Group Work	
January 30th 2018	Market Analysis by Anne McPhee	Guest speaker
February 1st 2018	NO CLASS	
February 6th 2018	First presentation_Morning	
February 8th 2018	First presentation_Afternoon	
February 13th 2018		
February 15th 2018		
February 20th 2018		
February 22nd 2018		
February 27th 2018	Presentation 2 Guidelines by Dr. Bilgin	
March 1st 2018	NO CLASS	
March 6th 2018	Second presentation_Afternoon	
March 8th 2018	Second presentation_Morning	
March 13th 2018		
March 15th 2018		
March 20th 2018		
March 22nd 2018	How to prepare scientific poster by Dr. Bilgin	
March 27st 2018	NO CLASS	
March 29rd 2018	NO CLASS	
April 3rd 2018	Third presentation	
April 5th 2018	Third presentation	
April 10th 2018		
April 12th 2018		
April 17th 2018		
April 19th 2018		
April 24th 2018		
April 26th 2018	NO CLASS	
April 27th 2018	EXPO	
May 1st 2018	Final presentation	
May 3rd 2018	Final presentation	

Exams:

There will be not be any exam.

Attendance:

Students are expected to attend all classes, and attendance is incorporated as 5% of the final grade. Students are allowed two excused absences or one unexcused absence. If absences exceed these limits, the student will receive a zero for attendance.

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance: 5%
Presentation 1: 10%
Report 1: 10%
Presentation 2: 10%
Report 2: 10%
Presentation 3: 10%
Report 3: 10%
Final Presentation: 12.5%
Final Report: 12.5%
Meeting with Mentors:............................. 10%